

Cheng Yang

PHONE NUMBER: (+1)7735718942 | EMAIL: chengya@uchicago.edu

EDUCATION

WUHAN UNIVERSITY

Bachelor in Information Systems | Average Score: 88.8/100

University of California, Berkeley

Berkeley Global Access Visiting Student Program

UNIVERSITY OF CHICAGO

Master in Computer Science

Sep. 2021 – Jun. 2025

Wuhan, China

Jan. 2024 – Aug. 2024

Berkeley, CA, USA

Sep. 2025 – At present

Chicago, IL, USA

PUBLICATIONS

- **C. Yang**, C. Shi, L. Jiang. PAVE: Prompt-Aligned Video Exploration via Training-Free Inpainting-to-Video World Model. Under review at AAAI 2027.
- **C. Yang**, W. Gao, D. Forsyth, A. Bhattad, M. Maire. Semantic and Mechanics Foam Fields for Physics-aware Editing. Under review at 3DV 2027.

RESEARCH EXPERIENCE

Compositional Visual Conditioning for Long-Horizon Robot Manipulation

Research Assistant, supervised by Prof. Jiatao Gu

Jul. 2026 – At present

UPenn, Pennsylvania, PA

- Investigating how pretrained visual world and action models can generalize from short-horizon demonstrations to long-horizon manipulation tasks that require composing multiple skills across changing scenes and intermediate states.
- Developing methods to retrieve, adapt, and compose task-relevant visual and action context from heterogeneous demonstrations while preserving the spatial and temporal structure necessary for downstream execution.
- Studying scene-aware conditioning and intermediate world-state representations as interfaces between visual prediction, high-level task composition, and closed-loop robotic control.

Structured Radiance Fields and Mechanics-Aware 3D Editing

Research Assistant, supervised by Prof. Michael Maire

Sep. 2025 – At present

UChicago, Chicago, IL

- Developed a structured radiance-field representation that models a scene using a compact set of ellipsoidal Gaussian cells, each parameterizing an independent local NeRF, enabling object- and part-centric scene representations with strong spatial locality.
- Derived a hybrid volume-rendering formulation that combines per-cell ray intersections and opacity integration with global depth-sorted compositing, allowing multiple local radiance fields to render jointly while preserving primitive-level ownership.
- Introduced a label-free hierarchical decomposition pipeline that voxelizes learned density fields, constructs graphs from geometry and DINO features, and applies spectral clustering with density distillation to recursively discover semantically meaningful components.
- Extended the representation with a mechanics-aware deformation framework that propagates user edits through cell connectivity, material, gravity, and contact constraints, then transfers the optimized deformation back to the underlying local radiance fields.

Active 3D World Expansion with Persistent 3D Memory

Research Assistant, supervised by Prof. Li Jiang

Jun. 2025 – Jul. 2026

CUHK(sz), Shenzhen, China

- Developed a training-free recurrent world-expansion framework that couples video-based scene generation with streaming 3D reconstruction, enabling long-horizon exploration from a single image or incomplete 3D observation.
- Built a persistent point-to-Gaussian memory that accumulates geometric evidence from generated observations, promotes verified regions into a renderable Gaussian representation, and conditions subsequent generation on the evolving 3D world state.
- Proposed an active Next-Best-Segment planner that organizes exploration into transit, inspection, and repair stages, maintaining view overlap and limiting reconstruction drift within a closed-loop generation, mapping,

and rendering system.

Skeleton-based Emotion Recognition

Dec. 2024 – July. 2025

Research Assistant. supervised by Prof. Wenwu Yang

China

- Proposed PA-GCN, a novel graph convolutional network for skeleton-based emotion recognition that adaptively models per-input, dynamically evolving emotional joint dependencies.
- Introduced class-specific emotion prototypes informed by frequency-domain emotion priors to generate adaptive topological connectivity weights for each input, enhancing emotion representations.
- Validated approach with extensive experiments — consistently outperformed state-of-the-art methods on both controlled and in-the-wild benchmarks and demonstrated strong robustness to class imbalance.

PROFESSIONAL EXPERIENCE

NetEase

Oct. 2024 – Jan. 2025

Intern

Hangzhou, China

- Independently designed and developed an interactive campaign H5 website using JavaScript, including designing UI elements and implementing core functionalities
- Integrated the company's user database into the website using MySQL for seamless data access

HONORS AND OTHER EXPERIENCES

- **Provincial project of College Student Innovation and Entrepreneurship Competition** **Sept. 2023**

Awards for outstanding projects in college student scientific research competitions

- **Third Prize in National College Student Mathematics Competition** **Dec.2022**

Awards for outstanding performance in the National College Student Mathematics Competition

SKILLS

- **Tech Skills:** Python (Pytorch, Pandas, Numpy, Opencv), C, Javascript, Java, Html, C++, SQL